

AI will eat electricity for breakfast and all day.

Posted by u/FairEstablishment851 • 0 points • 12 comments

Minute quantities of electricity used by one computer to create AI images, are really not a problem. I believe the sheer number of AI users and the increased AI power needs of business, will drive the need for a lot more electricity. It is going to take massive amounts of computing power and a lot of supercomputers to handle all of the new AI interactions in the world and it is going to take massive amounts of electricity to power all of those AI interactions and supercomputers, as well.

Comments

u/ninjasaid13 • 2 points • 2025-01-01 18:54 UTC

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which isn't really that much compared to other scaled things like Netflix as a whole, hosting websites, sending emails, video game industry as a whole, etc. Nothing is unique in regards to image generation as a whole.

u/spitfire_pilot • 5 points • 2025-01-01 18:19 UTC

Why focus on AI when emotional support vehicles for fragile egos are far more deleterious. Driving a pickup truck for a simple commute is like using a chainsaw to shave. It's overkill.

How about focusing on our weird habit of maintaining unproductive monoculture of foreign grasses which need constant work? Lawns are the ultimate waste of resources.

Maybe we could limit the absolute waste of using bottled waters in areas with clean tap water? It's insanity that we all pay public utilities and then have it marred by people drinking out of disposable containers that house the same stuff that comes out of taps.

There are tons of arguments that could be leveled at society for it's terrible resource management. AI is not near the top of that list. If anything it'll reduce our energy cinsumto8 eventually by optimization and helping us create green technologies.

u/Tyler_Zoro • 9 points • 2025-01-01 17:23 UTC

US Energy consumption per capita grew most sharply in the 1930s through 1960s. In recent decades, the energy consumption per capita has been dropping, even as we've built datacenter after datacenter. There's no rational reason to think that that's going to change.

u/lovestruck90210 • 2 points • 2025-01-02 00:54 UTC

According to the [EIA](https://www.eia.gov/outlooks/steo/), US energy consumption has been trending upwards since 2021. We moved from 3945 billion kilowatthours in 2021, to 4067 in 2022. There was a slight dip in 2023 to 4012 and then it increased to 4086 billion kilowatt hours in 2024. This trend is expected to continue in 2025, to an estimated 4165 billion kilowatt hours.

u/Tyler_Zoro • 1 points • 2025-01-02 03:21 UTC

The page you linked to shows an extremely low rate of growth in those three years that would be commensurate with increases in residential electricity consumption (electric cars, conversion to electric heating, increased reliance on electric cooling, etc.) But over the entire period that large datacenter growth has been prevalent, as I pointed out above, the overall electricity consumption has been pretty much level (and this small increase doesn't really change that trend).

u/lovestruck90210 • 1 points • 2025-01-02 05:48 UTC

You're just wrong. Again, from the [IEA](https://www.eia.gov/energyexplained/electricity/use-of-electricity.php) (https://www.eia.gov/energyexplained/electricity/use-of-electricity.php):

\> Total U.S. electricity consumption in 2022 was about 4.07 trillion kWh, the highest amount recorded and 14 times greater than electricity use in 1950. Total annual U.S. electricity consumption increased in all but 11 years between 1950 and 2022, and 8 of the years with year-over-year decreases occurred after 2007.

Electricity consumption has been trending upwards. We're consuming more electricity in the US than we ever have in history, and I'm not really seeing anything in the almost 12 year old NPR article you cited to contradict that.

\> In recent decades, the energy consumption per capita has been dropping, even as we've built datacenter after datacenter.

Goldman-Sachs [\[estimates\]](https://www.goldmansachs.com/pdfs/insights/pages/generational-growth-ai-data-centers-and-the-coming-us-power-surge/report.pdf) (https://www.goldmansachs.com/pdfs/insights/pages/generational-growth-ai-data-centers-and-the-coming-us-power-surge/report.pdf) that data centers are expected to account for 8% of US electricity by 2028, with AI accounting for 19% of the electricity demands of these centers by 2028 as well.

\> Our base case implies data center power demand moves from 1%-2% of overall global power demand to 3%-4% by 2030. The increase in the US is even greater — from 3% to 8%. Our estimates for overall data center power demand are above IEA forecasts (2026), and our outlook for AI to represent about 19% of data center power demand in 2028 is above recent corporate forecast.

Companies like Google and Microsoft are even looking to build [\[nuclear](#)

reactors\])(<https://www.bbc.com/news/articles/c748gn94k95o>) to power their data centers in large part due to AI.

\> Google has signed a deal to use small nuclear reactors to generate the vast amounts of energy needed to power its artificial intelligence (AI) data centres.

Is this just poor reporting from the antis over at BBC? Nope. Here's what the senior director for Energy and Climate over at Google had to say:

\> "The grid needs new electricity sources to support AI technologies," said Michael Terrell, senior director for energy and climate at Google. "This agreement helps accelerate a new technology to meet energy needs cleanly and reliably, and unlock the full potential of AI for everyone."

u/Tyler_Zoro • 1 points • 2025-01-02 12:09 UTC

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Um... okay. So you just quoted the section that re-states what I said above. I think you need to re-read that.

1. Between 1950 and the 2000s, electricity consumption rose steadily. It didn't jump higher when we started building large numbers of datacenters. Why? Because, though a datacenter's consumption is high, it consolidates many services for large numbers of people.
2. In the 2000s and through to fairly recently, the line has been relatively flat (some up, some down, but pretty flat). Why? More or less, there's a physical constraint on power production with supply and demand causing increases in price as we approach our peak production.
3. In the last few years, there has been an increase in consumption that is relatively small compared to the previous growth, but which is measurable. This represents new sources of power coming on line. IMHO, the largest contributor to this is power storage (the ability to generate power during off periods when consumption is low and reclaim it during peak periods) through a large number of strategies such as batteries,

inertial storage, pumped storage, etc. But that's just my personal theory.

u/Dr-Mantis-Tobbogan • 6 points • 2025-01-01 17:38 UTC

>rational reason

Man, I wished the antis cared about those.

u/RuukotoPresents • 3 points • 2025-01-01 17:15 UTC

Old simulations required more computing power and energy, you can run AI interactions locally on your laptop, even without the new NPU chips. With the chips, it's extremely efficient.

u/Katana_sized_banana • 3 points • 2025-01-01 16:58 UTC

Now while creating and learning an AI model takes a lot of power, this is a process only required once per model. People on the other hand do also use up power by different means.

While this is in nowhere near perfect of a study, because of a few open points.

<https://www.nature.com/articles/s41598-024-54271-x>

It does point towards some realistic reasons, why power is saved. My personal biggest issue is the reduced work via AI, doesn't automatically mean those man-hours aren't spend doing other work, that still requires a computer. However this could mean less processing power, by doing less computational demanding work. By saving energy from using a computer mostly in low energy states (other project work) instead of rendering over and over (Creating an image with a photo editor, 3D editor, video rendering etc).

If AI replaces phone assistance for instance, one less employee is using heating, light and other resources. (leaving aside the big question of UBI obviously)

u/justanotherponut • 6 points • 2025-01-01 16:57 UTC

And what about all the gpus used for gaming purposes and not ai use?

u/gcpwnd • 8 points • 2025-01-01 16:51 UTC

You mean like people use AI and it uses electricity? Like a fridge? That's crazy.

Let's ban fridges.

u/yunghelsing • 0 points • 2025-01-01 17:54 UTC

the fridge has a purpose in stark contrast to ai waifu nudes

u/gcpwnd • 4 points • 2025-01-01 18:57 UTC

Did you ever try to jerk off looking at a fridge?